TECHFEST 2026 | DHARAMVEER SAMBHAJI VIDYALAY

Home Automation

using Arduino & Bluetooth

Smart Home | Manual and Smartphone Control

What We Will Cover

Why Home Automation?

Convenience

Control lights, fans & appliances from your phone — no need to walk to every switch.

Energy Saving

Turn off forgotten devices automatically — save electricity & reduce bills.

Security

Manage home devices remotely even when you're away from home.

Accessibility

Helps elderly and differently-abled people operate home appliances easily.

Arduino Uno — The Brain

- Open-source microcontroller board
- ATmega328P chip — 16 MHz clock speed
- 14 Digital I/O pins | 6 Analog pins
- Operates at 5V, easily powered via USB
- Programmable using Arduino IDE (C/C++)
- Controls relays, sensors, motors & more

Specification	Value
Microcontroller	ATmega328P
Operating Voltage	5V
Digital I/O Pins	14 (6 PWM)
Analog Pins	6
Flash Memory	32 KB
Clock Speed	16 MHz
USB Interface	Type-B

⚡ In our project, Arduino Uno reads Bluetooth commands and switches home appliances ON/OFF via relay modules.

HC-05 Bluetooth Module

Wireless Range

Up to 10 metres
Indoor line-of-sight

Communication

UART Serial
Baud: 9600 bps

Pairing

Master/Slave mode
Pairs with smartphone

HC-05 Pin	Connect To	Purpose
VCC	5V (Arduino)	Power supply
GND	GND (Arduino)	Ground
TXD	Pin 10 (RX)	Transmit data
RXD	Pin 11 (TX)	Receive data

Components & Circuit | 04

Components Required

Component	Quantity
Arduino Uno	1 unit
HC-05 Bluetooth	1 unit
4-Channel Relay	1 unit
LED / Bulb	4 units
9V Battery / Adapter	1 unit
Jumper Wires	As needed
Android Smartphone	1 unit
Breadboard	1 unit

Circuit Flow Diagram

Smartphone App

▼

HC-05 Bluetooth

▼

Arduino Uno

▼

Relay Module

▼

Appliances

Smartphone Control via Bluetooth

1. **Download App**
   Install 'Arduino Bluetooth Controller' or 'BT Voice Control' from Play Store

2. **Pair Bluetooth**
   Enable Bluetooth on phone → Pair with HC-05 (Password: 1234 or 0000)

3. **Connect & Open**
   Launch the app → Connect to HC-05 module

4. **Assign Commands**
   Map buttons: '1' = Light ON, '0' = Light OFF, '2' = Fan ON, etc.

5. **Control!**
   Tap buttons to toggle appliances wirelessly from smartphone

Two Modes of Operation

✋ Manual Mode

- Physical push buttons on control panel
- Directly wired to Arduino digital pins
- Arduino reads HIGH/LOW signal
- Relay switches the appliance circuit
- LED indicator shows current state
- Works even without Bluetooth/phone

VS

📱 Smartphone Mode

- Open Bluetooth controller app on phone
- Connect to HC-05 module (up to 10m)
- Send character command (e.g. '1', '0')
- HC-05 forwards to Arduino via serial
- Arduino activates/deactivates relay
- Appliance turns ON or OFF instantly

Sample Arduino Sketch

```
// Home Automation - Arduino Bluetooth Code
#define RELAY1 8  // Light
#define RELAY2 9  // Fan

void setup() {
  Serial.begin(9600);
  pinMode(RELAY1, OUTPUT);
  pinMode(RELAY2, OUTPUT);
}

void loop() {
  if (Serial.available()) {
    char cmd = Serial.read();
    if (cmd == '1') digitalWrite(RELAY1, LOW);  // Light ON
    if (cmd == '0') digitalWrite(RELAY1, HIGH); // Light OFF
    if (cmd == '2') digitalWrite(RELAY2, LOW);  // Fan ON
    if (cmd == '3') digitalWrite(RELAY2, HIGH); // Fan OFF
  }
}
```

Applications & Future Scope | 07

Real-World Applications

Smart Home
Control lights, fans, AC & TV automatically

Offices
Centralized building automation

Hospitals
Remote control for patient rooms

Disabled Support
Voice/phone control for mobility aid

Industries
Automated machinery control

Future Enhancements

Wi-Fi / IoT Control
Control from anywhere using internet

Voice Commands
Add Google Assistant / Alexa support

Energy Monitor
Track electricity usage in real-time

Smart Security
Add motion sensors & camera alerts

AI Automation
Auto-schedule based on usage patterns

Key Takeaways

Simple & Affordable
Built with low-cost, easily available components

Dual Control
Works both manually and via smartphone Bluetooth

Practical Learning
Covers electronics, coding, and IoT concepts

Energy Efficient
Reduces unnecessary power consumption

Expandable
Can be extended to full smart home with Wi-Fi/IoT

Thank You!

Home Automation using Arduino Bluetooth

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Questions are Welcome!

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